Date:13.01.2021

2020-2021 FALL SEMESTER HYDRAULICS COURSE FINAL EXAM

Name: **Surname:** **Student ID:**

Question 1 (30):

In the Figure below, the water surface elevation of reservoir A is highest while the water surface elevation of reservoir C is the lowest. If the discharge from point D to reservoir C is **Q=0.10c** m^3/s Determine the discharge at each pipe and the surface elevation of the reservoir A. Draw the energy grade line (EGL) and hydraulic grade line (HGL) of the system. ($g=9.81\ m/s^2$).

Important Note: All the lengths and elevations of the reservoirs in the Figure are in **meter**s. In Figure, L is the length of the pipe, D is the diameter of the pipe, n is Manning friction factor, C is the Chezy coefficient, k is the roughness height of the pipe and λ is the Darcy-Weisbach friction factor. Where c,b, and a are the last three digits of your student ID. For example, if the student ID is 1234**567**, **a is 5**, **b is 6** and **c is 7**.

